RESEARCH SHOWCASE / FOUR-PAGE SUMMARY

Conditional hybrid retrieval

The selected paper, our adaptation, the measured changes, and what remains

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1. Base paper

The selected paper, Blended RAG by Sawarkar, Mangal, and Solanki, combines keyword retrieval, dense vectors, and learned sparse retrieval in Elasticsearch. It uses ELSER for the learned sparse branch and evaluates both retrieval rankings and generated-answer quality [1]. The paper motivates our question: must every query run every retrieval branch?

> Our question: Can a BM25-first system skip neural retrieval on some queries while keeping recall close to a blended reference?

Dimension	Blended RAG paper	Our project
Search stack	Elasticsearch; keyword, dense, and ELSER retrieval	Python BM25, MiniLM, SPLADE, reciprocal rank fusion
Research focus	Hybrid retrieval and answer quality	Query cost under recall-retention targets
Quality measures	Top-k, nDCG@10, generated-answer EM/F1	Recall@100 and retention, nDCG@10, MAP@10, latency
Comparison	Published paper results	Within-project tests; no direct original-system timing

The project studies the execution cost of conditional retrieval in our implementation. The published values provide context, but they are not a matched baseline for our experiments.

RECREATION AND ADAPTATION

2. A related recipe, implemented differently

We recreated the broad retrieval setup on overlapping datasets, then tested a BM25-first cascade. This is a methodological reproduction, not an exact reproduction of the original Elasticsearch and ELSER system.

What carries over	What changes in our study
Combining lexical, dense, and learned-sparse retrieval signals.	Python backend, MiniLM, SPLADE, reciprocal rank fusion, and Best Fields BM25 replace the original stack.
TREC-COVID and SQuAD provide task overlap with the paper.	SQuAD uses paragraph documents from official data; indexed follow-up tests a fixed sample of 256 questions.
A blended retrieval result provides a within-project reference.	We report Recall@100, recall retention, ranking measures, and latency; generated-answer EM/F1 is not part of the completed evaluation.

The first cascade exposed a bottleneck

In the initial TREC-COVID run, BM25 took 1,410.14 ms out of 1,493.23 ms for full fusion. Nearly all measured time sat in the lexical first stage, so skipping later neural work could save little. This result motivated indexed BM25 before we assessed conditional routing.

> Metric note: Recall retention is the method's mean Recall@100 divided by the fusion reference's mean Recall@100. It does not measure result-list overlap, and it is not generated-answer accuracy.

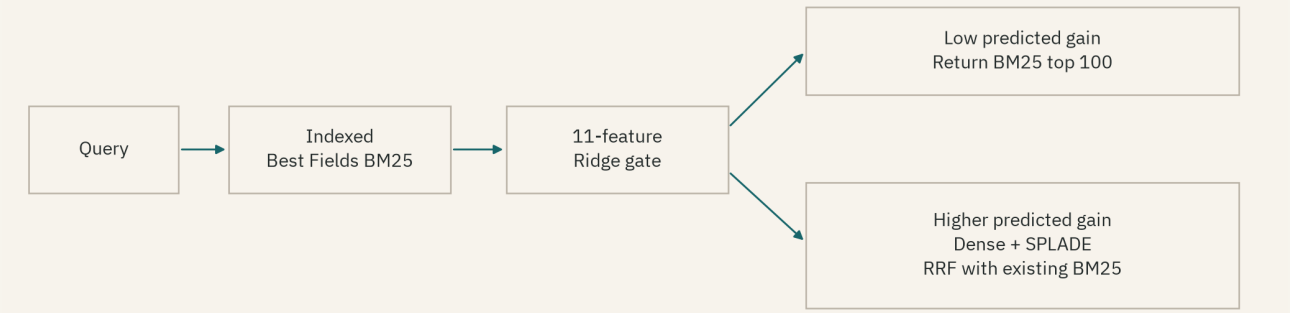
The indexed SQuAD result covers a declared 256-question sample against the full corpus. It is not a full-query evaluation.

CONTRIBUTION AND IMPLEMENTATION

3. Index the lexical stage, then decide whether to escalate

Our engineering contribution is an indexed BM25 implementation applied to fusion and cascade, plus a frozen BM25-first gate. We do not claim a new retrieval algorithm.

At inference, BM25 runs first. Eleven features from the query and BM25 ranking enter a standardized Ridge regression model trained on 2,590 NFCorpus training queries. It predicts the recall gain expected from neural retrieval. Low predicted gain returns BM25; otherwise MiniLM and SPLADE run, then RRF combines them with the BM25 ranking already available. The gate does not use relevance labels at inference.



Conditional path in the indexed cascade. Corpus and index preparation happen before query timing.

TREC-COVID / ALL 50 QUERIES

1.24x

Routing gain over indexed fusion

96.68% of indexed fusion mean Recall@100 retained.

20.21x combined

vs earlier Python fusion; includes indexing and routing. Indexed fusion alone was 16.36x.

SQUAD / FIXED SAMPLE OF 256 QUESTIONS

1.59x

Routing gain over indexed fusion

99.22% retained on the declared sample.

7.91x combined

vs earlier Python fusion; includes indexing and routing on the same sample.

The routing-only comparison uses equally indexed fusion. The combined ratios compare separate runs and must not be read as speedups against the original paper system.

FUTURE WORK / REFERENCES

4. What we learned, and what remains

The measurements show an engineering gain in our Python setup, not that the cascade is the best retrieval choice in every setting. Standalone SPLADE is a strong speed-recall option on TREC-COVID; on sampled SQuAD, standalone BM25 and SPLADE also provide competitive choices.

Limits of the current evidence

• System transfer: we did not evaluate the original Elasticsearch and ELSER pipeline.

• Evaluation scope: indexed SQuAD uses 256 queries; full NQ, HotpotQA, CoQA, and generated-answer quality remain pending.

• Selection and hardware: indexed TREC quality had already been inspected when its primary setting was selected; timings use one GPU type and few sessions.

• Practical costs: production concurrency, energy, index memory, and update costs were not measured.

Next study

Implement the cascade in Elasticsearch with ELSER and matched queries, corpus, and relevance judgments. Predeclare the protocol, test held-out collections, report generated-answer EM/F1, compare strong standalone and adaptive baselines, and measure repeated timing, load, memory, and index-update costs.

Summary: Indexed BM25 addressed the lexical bottleneck in our implementation, and routing added a measured gain over equally indexed fusion. Testing whether those gains transfer to the original system is the next step.

Selected references

[1] Sawarkar, Mangal, and Solanki (2024). Blended RAG: Improving RAG Accuracy with Semantic Search and Hybrid Query-Based Retrievers. arXiv:2404.07220.
[2] Arabzadeh, Yan, and Clarke (2021). Predicting Efficiency/Effectiveness Trade-offs for Dense vs. Sparse Retrieval Strategy Selection. arXiv:2109.10739.
[3] Lu (2024). BM25S: Orders of Magnitude Faster Lexical Search via Eager Sparse Scoring. arXiv:2407.03618.
[4] Thakur et al. (2021). BEIR: A Heterogeneous Benchmark for Zero-shot Evaluation of Information Retrieval Models. NeurIPS Datasets and Benchmarks.
[5] Rajpurkar et al. (2016). SQuAD: 100,000+ Questions for Machine Comprehension of Text. EMNLP.
[6] Formal, Piwowarski, and Clinchant (2021). SPLADE: Sparse Lexical and Expansion Model for First Stage Ranking. SIGIR.
[7] Reimers and Gurevych (2019). Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. EMNLP-IJCNLP.